Progressive exercises

E1 · Practice. Consider the sequence (u_n) defined, for every natural number n, by $u_n = 5 - 3n$. Calculate u_4, then the sum $\sum_{k=0}^{20} u_k$.

E2 · Practice. The sequence (v_n) is defined by $v_0 = 3$ and, for every natural number n, $v_{n+1} = 2v_n$. Express v_n in terms of n, then calculate $\sum_{k=0}^{n} v_k$.

E3 · Practice. Prove by induction that, for every natural number n, $3^n \geq 2n + 1$.

E4 · Practice. Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = \frac{n+1}{n+2}$. Prove that it is increasing and give bounds valid for all its terms.

E5 · Practice. Prove by induction that, for every natural number n, $8^n - 1$ is divisible by 7. Do not use congruences.

Textbook or past-paper work

Barbazo, p. 28, exercises 22 and 25. In class, in S5: exercise 22, to translate properties of sequences using a quantifier. At home after S11: the whole of exercise 25, to combine induction, upper bounds and monotonicity. Write the base case, the inductive step and the conclusion separately.